Mathematical Summary of the YQuantum 2026 Hartford Portfolio Project

1 Problem Setup

The project studies a reduced insurer portfolio selection problem built from real Hartford workbook data The original workbook contains 50 assets, 1200 economic scenarios, and an asset-level covariance matrix These 50 assets are aggregated into 8 sectors:

Cash, Gov Bonds, IG Credit, HY Credit, Equities US, Equities Intl, Real Estate, Infrastructure

The optimization problem is:

Out of 8 sector-level choices, which 4 sectors should an insurer hold to balance return, covariance risk, capital bu

This is a binary combinatorial problem Let

$$x = (x_1, \ldots, x_8) \in \{0,1\}^8,$$

where $x_i = 1$ means sector i is selected and $x_i = 0$ means it is not selected The project imposes the exact-cardinality condition

$$\sum_{i=1}^{8} x_i = B, \qquad B = 4$$

Thus the feasible set contains

$$\binom{8}{4} = 70$$

valid portfolios out of

$$2^8 = 256$$

possible bitstrings

2 Aggregation from Assets to Sectors

Let

$$N = 50, \qquad S = 1200, \qquad K = 8$$

Let

$$R \in \mathbb{R}^{S \times N}$$

be the asset-level scenario-return matrix The project defines an aggregation matrix

$$A \in \mathbb{R}^{K \times N}$$

by

$$A_{k,a} = \begin{cases} \dfrac{1}{|A_k|}, & \text{if asset } a \text{ belongs to sector } k, \\ 0, & \text{otherwise,} \end{cases}$$

where A_k is the set of assets in sector k

The sector-level scenario matrix is then

$$R_{\text{sec}} = RA^\top \in \mathbb{R}^{S \times K}$$

From this matrix, the project computes the expected sector returns

$$\mu = \frac{1}{S} \sum_{s=1}^{S} (R_{\text{sec}})_s \in \mathbb{R}^8$$

and the sector covariance matrix

$$\Sigma = \text{Cov}(R_{\text{sec}}) \in \mathbb{R}^{8 \times 8}$$

This design is intentionally scenario-driven: the scenario sheet determines expected return, covariance, worst-case return, VaR, and CVaR

3 Capital and Liquidity Inputs

For each sector k, the project averages the asset-level capital charges and liquidity scores:

$$c_k = \frac{1}{|A_k|} \sum_{a \in A_k} c(a), \qquad \ell_k = \frac{1}{|A_k|} \sum_{a \in A_k} \ell(a)$$

Capital is min-max normalized:

$$\tilde{c}_k = \frac{c_k - \min(c)}{\max(c) - \min(c)}$$

Liquidity is normalized and then converted to illiquidity:

$$\tilde{\ell}_k = \frac{\ell_k - \min(\ell)}{\max(\ell) - \min(\ell)}, \qquad \tilde{i}_k = 1 - \tilde{\ell}_k$$

Thus larger $\tilde{i}_k$ means the sector is less liquid and therefore receives a larger penalty

4 Portfolio Metrics

Given a binary selection vector $x \in \{0,1\}^8$ with target budget $B = 4$, the project uses equal weights

$$w = \frac{x}{B}$$

The resulting portfolio metrics are:

$$R(x) = \frac{1}{B} \mu^\top x$$

for expected return,

$$V(x) = \frac{1}{B^2} x^\top \Sigma x$$

for variance, and

$$\sigma(x) = \sqrt{V(x)}$$

for volatility

The project also computes scenario-level portfolio returns For scenario s,

$$r^{(s)}(x) = (R_{\text{sec}})_s \cdot \frac{x}{B}$$

From these 1200 scenario returns, it computes:

- worst scenario return,
- Value-at-Risk at level $\alpha = 0.95$,
- Conditional Value-at-Risk at level $\alpha = 0.95$

Returns are converted to losses by

$$\text{loss} = -\text{return},$$

and then

$$\text{VaR}_\alpha = \text{Quantile}_\alpha(\text{loss}), \qquad \text{CVaR}_\alpha = \mathbb{E}[\text{loss} \mid \text{loss} \geq \text{VaR}_\alpha]$$

The project also tracks descriptive insurance metrics:

$$C(x) = \frac{1}{B} c^\top x, \qquad L(x) = \frac{1}{B} \ell^\top x$$

5 Insurer-Aware QUBO Formulation

The base Markowitz-style binary objective is

$$\min_{x \in \{0,1\}^8} \frac{q}{B^2} x^\top \Sigma x - \frac{1}{B} \mu^\top x + \lambda (1^\top x - B)^2$$

To make the model insurance-specific, the project adds capital and illiquidity penalties:

$$\min_{x \in \{0,1\}^8} \frac{q}{B^2} x^\top \Sigma x - \frac{1}{B} \mu^\top x + \lambda (1^\top x - B)^2 + \frac{\gamma}{B} \tilde{c}^\top x + \frac{\eta}{B} \tilde{i}^\top x$$

The default hyperparameters are:

$$B = 4, \qquad q = 0.5, \qquad \lambda = 8.0, \qquad \gamma = 0.25, \qquad \eta = 0.25$$

The four model variants considered are:

Variant	γ	η
base	0	0
capital	025	0
liquidity	0	025
both	025	025

6 Expanding the Budget Penalty

The exact-cardinality constraint is enforced through the soft penalty

$$\lambda(1^\top x - B)^2 = \lambda\left(\sum_i x_i - B\right)^2$$

Because $x_i^2 = x_i$ for binary variables,

$$\lambda\left(\sum_i x_i - B\right)^2 = \lambda\left(\sum_i x_i + 2\sum_{i<j} x_i x_j - 2B\sum_i x_i + B^2\right)$$

Therefore

$$\lambda\left(\sum_i x_i - B\right)^2 = \lambda\sum_i (1-2B)x_i + 2\lambda\sum_{i<j} x_i x_j + \lambda B^2$$

The constant term λB^2 does not affect the minimizer and is dropped in the QUBO representation

7 Upper-Triangular QUBO Matrix

The QUBO is stored as an upper-triangular matrix Q such that

$$f(x) = x^\top Q x$$

Define the scaling constants

$$a = \frac{q}{B^2}, \qquad b = \frac{1}{B}, \qquad g = \frac{\gamma}{B}, \qquad e = \frac{\eta}{B}$$

Then the diagonal entries are

$$Q_{ii} = a\Sigma_{ii} - b\mu_i + \lambda(1-2B) + g\tilde{c}_i + e\tilde{i}_i,$$

and the off-diagonal entries are

$$Q_{ij} = 2a\Sigma_{ij} + 2\lambda, \qquad i < j$$

The diagonal term Q_{ii} collects the one-sector contributions: sector variance, expected return, the one-body part of the budget penalty, capital penalty, and illiquidity penalty

The off-diagonal term Q_{ij} collects the two-sector interactions: covariance plus the pairwise part of the budget penalty

A key structural consequence is that because $\lambda > 0$, every off-diagonal entry receives the additive term 2λ Hence every pair of sectors is coupled, even if Σ_{ij} is small The interaction graph is therefore the complete graph

$$K_8,$$

which has

$$\binom{8}{2} = 28$$

edges

8 QUBO Scaling

Before converting to an Ising Hamiltonian, the code rescales the QUBO:

$$Q_{\text{scaled}} = \frac{Q}{\alpha}, \qquad \alpha = \max_{i,j} |Q_{ij}|$$

This keeps all entries in approximately $[-1, 1]$ and improves numerical conditioning for simulation and variational angle optimization

9 QUBO-to-Ising Mapping

To use QAOA, the binary QUBO must be rewritten in spin form The project uses the standard substitution

$$x_i = \frac{1 - s_i}{2}, \qquad s_i \in \{+1, -1\}$$

Equivalently,

$$x_i = 0 \iff s_i = +1, \qquad x_i = 1 \iff s_i = -1$$

After substitution, the cost Hamiltonian takes the Ising form

$$H_C = \sum_{i=1}^{8} h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j + \text{offset}$$

For an upper-triangular QUBO matrix Q, the coefficients are

$$J_{ij} = \frac{Q_{ij}}{4}, \qquad i < j,$$

and

$$h_i = -\frac{Q_{ii}}{2} - \frac{1}{4} \sum_{j \neq i} Q_{\min(i,j),\max(i,j)}$$

The offset is

$$\text{offset} = \sum_i \frac{Q_{ii}}{2} + \sum_{i<j} \frac{Q_{ij}}{4}$$

Thus the Hamiltonian is not chosen arbitrarily It is derived directly from the insurer-aware QUBO

10 QAOA Formulation

QAOA alternates the cost Hamiltonian

$$H_C = \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j$$

with the standard transverse-field mixer

$$H_M = \sum_i X_i$$

At depth p, the variational state is

$$|\gamma, \beta\rangle = \prod_{k=1}^{p} e^{-i\beta_k H_M} e^{-i\gamma_k H_C} |+\rangle^{\otimes 8}$$

The QAOA objective is the expected cost energy

$$\mathbb{E}[H_C] = \sum_z |\psi_z|^2 E(z),$$

where $E(z)$ is the precomputed Ising energy of basis state z

Because the system has only 8 qubits, the Hilbert space has dimension

$$2^8 = 256,$$

so exact statevector simulation is possible The project uses multi-start angle search followed by coordinate descent refinement

11 Exact Classical Baselines

Since the problem size is small, the project can evaluate

$$f(x) = x^\top Q x$$

for all

$$x \in \{0, 1\}^8$$

This gives:

- the best overall bitstring,
- the best feasible bitstring,
- a complete exact benchmark for QAOA

It also includes:

- a greedy baseline that adds one sector at a time,
- a continuous Markowitz proxy over the simplex

12 Exact Optimal Portfolios

The validated exact optima are:

Model	Bitstring	Selected sectors
base	00011110	HY Credit, Equities US, Equities Intl, Real Estate
capital	11110000	Cash, Gov Bonds, IG Credit, HY Credit
liquidity	11001100	Cash, Gov Bonds, Equities US, Equities Intl
both	11110000	Cash, Gov Bonds, IG Credit, HY Credit

This shows that adding a capital penalty rotates the optimum strongly into fixed income, while the liquidity penalty produces a middle-ground solution The fact that the "both" model matches the "capital" model indicates that, at the chosen weights, capital is the binding insurer constraint

13 Mathematical Interpretation of the Validated Quantum Results

The most important quantitative fact from the validated QAOA runs is:

$$\frac{P(\text{optimal})}{P(\text{feasible})} \approx \frac{1}{70}$$

Since there are exactly 70 feasible 4-of-8 portfolios, this means that conditional on feasibility, the optimal portfolio is sampled with probability essentially indistinguishable from the uniform distribution over the feasible set

At the same time, the project finds

$$P(\text{feasible}) \approx 05766 \gg \frac{70}{256} \approx 02734$$

So the correct mathematical interpretation is:

- QAOA learns the cardinality structure imposed by the penalty,
- but within the feasible set, shallow QAOA does *not* strongly distinguish the best portfolio from the other feasible portfolios

This is why the project claims correctness and meaningful hardware analysis, but does not claim quantum advantage on the 8-qubit instance